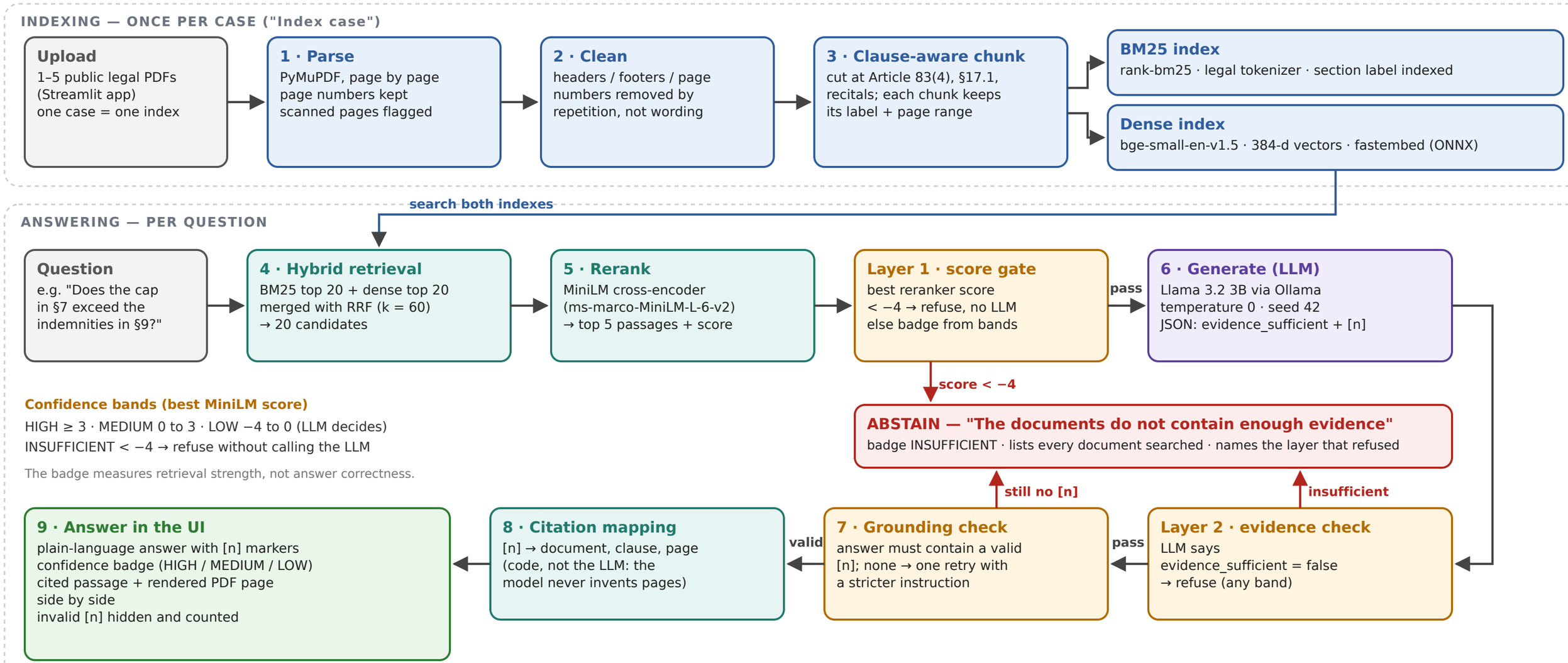
# LexiSyncAI — evidence-grounded legal document analyst

Hybrid retrieval + reranking, page/clause citations, two-layer abstention · fully local (Intel Mac, CPU only, Ollama)




## Evaluation · 20 held-out questions (12 factual, 4 multi-hop, 4 unanswerable), written before the code

| Recall@5 (retrieval) | Answer correctness | Citation precision | Abstention | Latency (fixed conditions) |
|---|---|---|---|---|
| **dense 0.88 → hybrid 0.97** | **11/16 correct** | **17/21 = 0.81** | **4/4 unanswerable refused** | **8.6 s mean / 9.1 s median** |
| → + MiniLM 1.00 | factual 10/12 · multi-hop 1/4 | cited passage supports claim | 0/16 answerable refused | retrieval ≈ 0.8 s, rest = LLM |

Mandatory comparison: dense-only vs hybrid. Known weakness: multi-hop answers (the 3B model often uses only one of two passages). github.com/mdl-erfan/LexiSyncAI